KENDRIYA VIDYALAYA SANGATHAN BHOPAL REGION

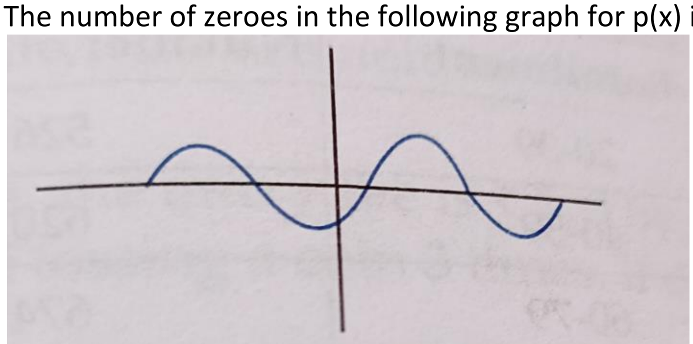
REGIONAL LEVEL PRE-BOARD I SET – 3

TIME : - 3 HOURS CLASS : - X SUBJECT : - MATHEMATICS M.M. 80

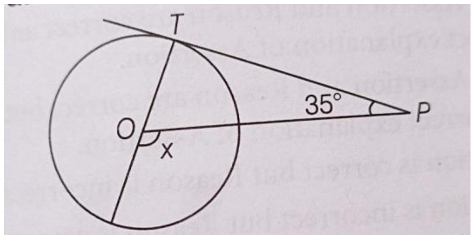
General Instructions :

Read the following instructions carefully and follow them :

1. *This question paper contains 38 questions.*
2. *This Question Paper is divided into 5 Sections A, B, C, D, and E.*
3. *In Section A, Questions no. 1-18 are Multiple Choice Questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.*
4. *In Section B, Questions no. 21-25are very short answer (VSA) type questions, carrying 2 marks each.*
5. *In Section C, Questions no.26-31are short answer (SA) type questions, carrying 3 marks each.*
6. *In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 5 marks each.*
7. *In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub-parts of the values of 1, 1 and 2 marks each respectively.*
8. *All Questions are compulsory. However, an internal choice in 2 Questions of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in the 2 marks questions of Section E.*
9. *Draw neat figures wherever required.*
10. *Take π =22/7 wherever required if not stated.*
11. *Use of calculators is not allowed.*

	SECTION A	
1.	From a solid cube of side 14 cm, a sphere of maximum diameter is carved out. The radius of sphere is (a)7 cm (b) 14 cm (c) $\frac{7}{2}$ cm (d) $\sqrt{14}$ cm	1
2.	The sum of the first 50 odd natural numbers is (a)5000 (b) 2500 (c) 2550 (d) 5050	1
3.	Two coins tossed together. The probability of getting exactly one tail is (a)$\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1	1
4.	The pair of linear equations, x = y and x – 2 = y – 2 are (a)perpendicular (b) parallel (c) intersecting (d) coincident	1
5.	The number of zeroes in the following graph for p(x) is  (a)0 (b) 3 (c) 5 (d) 1	1
6.	The discriminant of quadratic equation $x^2 - 4x + 1 = 0$ is (a)12 (b) 1 (c) 0 (d) 6	1

7. In the given figure, if PT is a tangent to a circle with centre O and $\angle TPO = 35^0$, then the measure of $\angle x$ is — 1



(a) 110^0 (b) 115^0 (c) 120^0 (d) 125^0

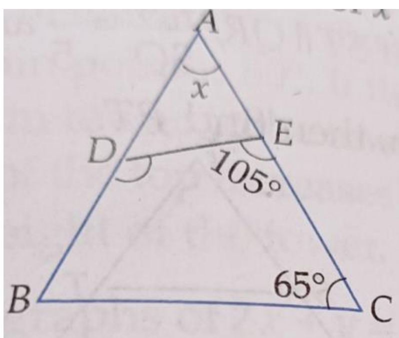
8. If the mean and median of a data are 10 and 11 respectively, then mode of the data is — 1

(a) 20 (b) 12 (c) 13 (d) 8

9. A quadratic polynomial having sum and product of its zeroes are 5 and 0 respectively, is — 1

(a) $x^2 + 5x$ (b) $x(x - 5)$ (c) $5x^2 - 1$ (d) $x^2 - 5x + 5$

10. If $\Delta ADE \sim \Delta ACB$, $\angle DEC = 105^0$ and $\angle ECB = 65^0$, then the value of x is — 1



(a) 60^0 (b) 90^0 (c) 30^0 (d) 40^0

11. $\frac{\sin^2\theta}{\cos^2\theta} - \frac{1}{\cos^2\theta}$ in simplified form, is — 1

(a) $\cot^2\theta$ (b) 1 (c) $\tan^2\theta$ (d) - 1

12. Two AP's have the same common difference. The first term of one of these is – 1 and that of the other is – 8. Then, the difference between their 4th term is — 1

(a) 1 (b) 8 (c) 7 (d) 9

13. The time in seconds, taken by 150 athletes to run a 100 m hurdle race are tabulated below — 1

Time (sec)	13-14	14-15	15-16	16-17	17-18	18-19
Number of Athletes	2	4	5	71	48	20

The number of athletes who completed the race in less than 17 seconds, is

(a) 82 (b) 71 (c) 68 (d) 11

14. The volume of a solid hemisphere is $\frac{2156}{3}$ cm^3. The total surface area of the solid hemisphere is — 1

(a) 308 cm^2 (b) 462 cm^2 (c) 803 cm^2 (d) 264 cm^2

15. The point of intersection of the line represented by $3x - y = 3$ and Y-axis is given by — 1

(a) (0, - 3) (b) (0, 3) (c) (2, 0) (d) (– 2, 0)

16.	A building and tower of height 75 m and 150 m stand vertically upright on a plane ground. If the distance between their feet is 75 m, then angle of elevation to tower from top of the building is (a)60^0 (b) 30^0 (c) 45^0 (d) 90^0	1
17.	Two concentric circles are of radii 10 cm and 8 cm, then the length of the chord of the larger circle which touches the smaller circle is (a)6 cm (b) 12 cm (c) 18 cm (d) 9 cm	1
18.	The point that divides the line segment joining the points (7, - 6) and (3, 4) in ratio 1 : 2 internally lies in which quadrant? (a)I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant	1
	DIRECTION: In the question number 19 and 20, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**. Choose the correct option A)Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) B)Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) C)Assertion (A) is true but reason (R) is false. D)Assertion (A) is false but reason (R) is true.	
19.	**Assertion(A)** The HCF of two numbers is 5 and their product is 150. Then, their LCM is 40. **Reason(R)** For any two positive integers a and b, HCF(a, b) × LCM(a, b) = a × b	1
20.	**Assertion(A)** If the area of sector is $\frac{6}{18}$ of the area of circle, then angle subtended by it at centre is 120^0 . Reason(R) Area of sector = $\frac{\theta}{360^0}$ × area of circle.	1
	SECTION B	
21.	If the distance between the points A(x, 2) and B(9, 8) is 10 units, then find the value(s) of x.	2
22.	Find the ratio in which the line segment joining the points P(6, 3) and Q(-2, - 5) is divided by X-axis. OR Find the perimeter of the triangle formed by points (0, 0), (1, 0) and (0, 1)	2
23.	The traffic lights at three different road crossings change after every 48 s, 72 s and 108 s, respectively. If they change simultaneously at 7:00 am, at what time will they change together next.	2
24.	Evaluate : $\frac{\sin 30^0 + tan45^0}{sec30^0 + cot45^0}$	2
25.	The angles of a triangle are in A. P. The greatest angle is twice the least. Find all the angles of the triangle. OR How many multiples of 9 lie between 10 and 300?	2
	SECTION C	
26.	Prove that $\sqrt{7}$ is an irrational number.	3

27. Prove that : **3**

$$\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} = 2 \sec A$$

OR

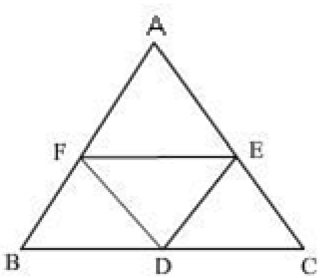
$$(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$$

28. A chord of a circle of radius 21 cm subtends an angle 60^0 at the centre. Find the area of the corresponding minor segment of the circle. [Use π = 3.14 and $\sqrt{3}$ = 1.73] **3**

29. If zeroes of the polynomial $x^2 + (a + 1)x + b$ are 2 and – 3, then find the value of (a + b). **3**

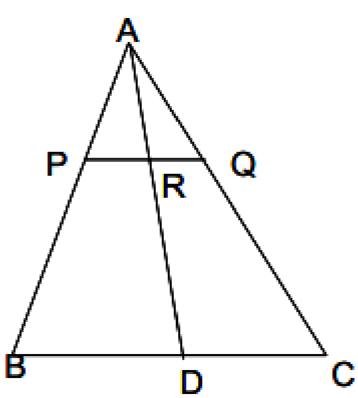
30. The sum of two numbers is 11 and the sum of their reciprocals is $\frac{11}{28}$. Find the numbers. **3**

31. In ΔABC, D, E and F are midpoints of BC, CA and AB respectively. Prove that $\triangle FBD \sim \triangle DEF$ and $\triangle DEF \sim \triangle ABC$. **3**



OR

In ΔABC, P and Q are points on AB and AC respectively such that PQ is parallel to BC. Prove that the median AD drawn from A on BC bisects PQ.



SECTION D

32. Find the mean and median of the following data: **5**

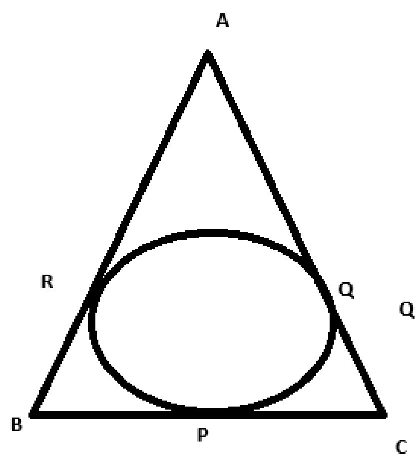
Class	40-45	45-50	50-55	55-60	60-65	65-70
Frequency	8	9	10	9	5	4

OR

The monthly expenditure on milk in 200 families of a Housing Society is given below

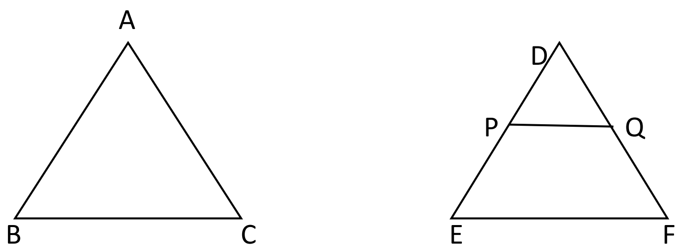
Monthly Expenditure (in Rs.)	1000-1500	1500-2000	2000-2500	2500-3000	3000-3500	3500-4000	4000-4500	4500-5000
Number of families	24	40	33	x	30	22	16	7

Find the value of x and also find the mean expenditure.

33.	From the top of a 20 m high building, the angle of elevation of the top of a cable tower is 60^0 and angle of depression of its foot is 45^0. Determine the height of the tower and distance between building and tower. (Use $\sqrt{3} = 1.732$) OR As observed from the top of a 100m high lighthouse from the sea level, the angles of depression of two ships are 30^0 and 45^0. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships (Use $\sqrt{3} = 1.732$)	5
34.	Prove that the lengths of tangents drawn from an external point to a circle are equal. Using above result, find the length BC of ΔABC. Given that, a circle is inscribed in ΔABC touching the sides AB, BC and CA at R, P and Q respectively and AB= 15 cm, AQ= 12cm, CQ=7cm  	5
35.	Solve the following system of linear equations graphically: $3x - 4y + 3 = 0, \quad 3x + 4y - 21 = 0$	5
	SECTION E	
36.	Metallic silos are used by farmers for storing grains. Farmer Ramu has decided to build a new metallic silo to store his harvested grains. It is in the shape of a cylinder mounted by a cone. Dimensions of the conical part of a silo is as follows: Radius of base = 6 m Height = 8 m Dimensions of the cylindrical part of a silo is as follows: Radius = 6 m Height = 14 m On the basis of the above information answer the following questions : (i)Calculate the slant height of the conical part of one silo. (ii) Find the curved surface area of the conical part of one silo. (iii)(A) Find the cost of metal sheet used to make the curved cylindrical part of 1 silo at the rate of ₹1000 per m^2 . OR (iii) (B) Find the total capacity of one silo to store grains.	1 1 2
37.	A group of students conducted a survey to find out about the preferred mode of transportation to school among their classmates. They surveyed 400 students from their school. The results of the survey are as follows: 180 students preferred to walk to school. 25% of the students preferred to use bicycles. 10% of the students preferred to take the bus. Remaining students preferred to be dropped off by car. Based on the above information, answer the following questions: (i)What is the probability that a randomly selected student does not prefer to walk to school? (ii) Find the probability of a randomly selected student who prefers to walk or use a bicycle. (iii) (**A**) One day 50% of walking students decided to come by bicycle. What is the probability that a randomly selected student comes to school using a bicycle on that day?	1 1 2

OR (iii)(**B)**What is the probability that a randomly selected student prefers to be dropped off by car?

38.



Triangle is a very popular shape used in interior designing. The picture given above shows a cabinet designed by a famous interior designer.

Here the largest triangle is represented by △ ABC and smallest one with shelf is represented by △ DEF. PQ is parallel to EF.

(i) Show that △ DPQ ~ △ DEF. 1

(ii) If DP= 100 cm and PE = 140 cm then find $\frac{PQ}{EF}$. 1

(iii) (A) If 2AB = 5DE and △ ABC ~ △ DEF then show that $\frac{perimeter\ of\ \Delta ABC}{perimeter\ of\ \Delta DEF}$ is constant. 2

OR (iii) (B) If AM and DN are medians of triangles ABC and DEF respectively then prove that △ ABM ~ △ DEN.